

NCAOS-Urielle EGS · Phase 03

Detection Layer — MTTD Instrumentation · @ncaos/detection v0.3.0 · Week 4–6

@ncaos/core	47/47	@ncaos/shell	15/15	@ncaos/api	8/8	@ncaos/detection	54/54
Cumulative	124/124	MTTD tests	12	Gate tests	18	Detection tests	24

@ncaos/api · Phase 02 — unchanged · 8 tests ✓
POST /v1/process · GET /v1/state · GET /v1/audit/:handle · GET /v1/health · WS /v1/state/stream

@ncaos/shell · Phase 02 — unchanged · 15 tests ✓
ShellLoop.process() · Observer→Judge→Enforcer · Watchdog 3-restart · ADR-003 process isolation

@ncaos/detection · v0.3.0 · NEW · 54 tests · ADR-004: detection layer independence
Each detector is a class, not a function. MTTD measured inside each detector. Rule sets configurable at construction.

Module	File	Capability	Tests
MTTD Tracker	mttd/tracker.ts	Ring buffer (1000 samples) · p50/p95/p99 stats · SLA breach detection · DetectionRegistry per policy · isSlaHealthy(): true if all layers <5% breach rate	12 tests

Detector	File	Key capabilities	Tests
GateDetector	gate/gate-detector.ts	10 configurable rules: 5 critical (INJECTION_PATTERN, SIGNATURE_INVALID, KNOWN_BAD_SOURCE, REPLAY_ATTACK, TAMPERED_PAYLOAD) → HIGH/sys. 5 warning (RATE_EXCEEDED, GEO_MISMATCH, SCHEMA_VIOLATION ...) → MED. Source allowlist · addRule() custom injection · getRules()	18 tests ✓
PremiseDetector	premise/premise-detector.ts	Weighted premises: critical/standard/info tiers. Expiry windows per premise. Per-workflow PremiseRegistry. Contract version mismatch detection. Unknown claim flagging.	9 tests ✓
AuthorityDetector	authority/authority-detector.ts	Full state machine: GRANT/DOWNGRADE/DENY/ESCALATION. Chain constraint: authority can only downgrade in a chain, never upgrade. Immutable AuthorityAuditLog. escalation_attempt → HIGH/sys. D-01 fix: escalation correctly HIGH.	9 tests ✓
ContinuityDetector	continuity/continuity-detector.ts	Weighted equivalence scoring. Change window FSM: OPEN→WATCHING→CLOSED→CONTAINMENT. Trend tracking. 12-test template generator. D-02 fix: OPEN→WATCHING is first transition (not CLOSED).	6 tests ✓

Gate detection rules — 10 configurable, 5 critical + 5 warning

Rule ID	Flag	Severity	Scope	Type
G-C-001	INJECTION_PATTERN	HIGH	sys	Critical

G-C-002	SIGNATURE_INVALID	HIGH	sys	Critical
G-C-003	KNOWN_BAD_SOURCE	HIGH	sys	Critical
G-C-004	REPLAY_ATTACK	HIGH	sys	Critical
G-C-005	TAMPERED_PAYLOAD	HIGH	sys	Critical
G-W-001	RATE_EXCEEDED	MED	req	Warning
G-W-002	UNRECOGNIZED_SOURCE	MED	req	Warning
G-W-003	GEO_MISMATCH	MED	req	Warning
G-W-004	SCHEMA_VIOLATION	MED	flow	Warning
G-W-005	OVERSIZED_PAYLOAD	MED	req	Warning

Phase 01 → Phase 03: function to class upgrade

Capability	Phase 01 (function)	Phase 03 (class — upgrade)
Gate	detectGate() — stateless, hardcoded flags, no MTTD	GateDetector — 10 configurable rules, source allowlist, MTTD measured, addRule() for custom injection
Premise	detectPremise() — ratio-based, no registry	PremiseDetector — weighted premises, expiry windows, per-workflow PremiseRegistry, contract version checks
Authority	detectAuthority() — simple override flag	AuthorityDetector — full FSM, chain constraint, immutable AuthorityAuditLog, escalation classification
Continuity	detectContinuity() — simple score threshold	ContinuityDetector — weighted scoring, OPEN→WATCHING→CLOSED→CONTAINMENT FSM, trend tracking
MTTD	mttdMs field present, not tracked	MttTracker: 1000-sample ring buffer, p50/p95/p99, SLA breach detection, isSlaHealthy()

Defects found & fixed — 3 issues, all resolved

ID	Test	Root cause	Fix applied
D-01	AuthorityDetector — downgrade severity	Requesting decision-ready when policy grants only reference-only is escalation_attempt → correctly HIGH. Test said 'med'.	Changed: expect(result.severity).toBe('high') — test alignment, not runtime bug
D-02	ContinuityDetector — window transition	OPEN→WATCHING is first transition. WATCHING→CLOSED requires subsequent stable check. Test said 'CLOSED'.	Changed: expect(result.changeWindowState).toBe('WATCHING')
D-03	File save path mismatch	VS Code opened from different folder. Edited file not the file npm test was reading.	Set-Content to correct PowerShell path. Going forward: code . from eggs/ directory.

Cumulative test results — 124 / 124 passing · 4 packages · 8 test files

Package	File	Suite / scope	Tests	Status
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@ncaos/core	scoring.test.ts	computeIntegrity + computeAuthority + metrics + continuity	16	
	detection.test.ts	detectGate + detectPremise + detectAuthority + detectContinuity	16	
	verdict.test.ts	enforce (all profiles) + failSafeVerdict + policy validation	15	47 ✓
@ncaos/shell	shell-loop.test.ts	lifecycle + clean PILOT + threat STRICT + event emission	15	15 ✓
@ncaos/api	api.test.ts	health + process (401/400/200/EVD) + state + audit	8	8 ✓
@ncaos/detection	mttd.test.ts	MttdTracker stats + breach + registry + startTimer	12	
	gate.test.ts	clean + 5 critical flags + warning flags + allowlist + custom	18	
	detection.test.ts	PremiseDetector + AuthorityDetector + ContinuityDetector	24	54 ✓
CUMULATIVE TOTAL	Phases 01-03	4 packages · 8 test files	124	ALL PASS ✓

Architecture decision records — ADR-001 through ADR-004

ADR	Phase	Decision	Consequence
ADR-001	Phase 01	Zod-first type contracts	All packages import from @ncaos/core. Breaking changes = version bump. New fields must be .optional().
ADR-002	Phase 01	Fail-safe default-deny	failSafeVerdict() cannot be disabled by policy. Shell failure → always BLOCK. Chaos test Phase 06.
ADR-003	Phase 02	Process isolation	ShellLoop as separate process. Watchdog 3-restart + exponential backoff. Permanent FAIL_SAFE on exhaustion.
ADR-004	Phase 03	Detection layer independence	Each detector is a class, not a function. MTTD measured inside. Rule sets configurable. Per-tenant in Phase 04.

Phase roadmap

Phase 01	Foundation & repo setup	W1-2	47/47	COMPLETE
Phase 02	Core shell engine + API skeleton	W2-4	70/70	COMPLETE
Phase 03	Detection layer — MTTD instrumentation	W4-6	124/124	COMPLETE
Phase 04	API integration surface	W6-8	TBD	NEXT
Phase 05	Admin UI (production)	W8-10	TBD	PLANNED
Phase 06	Hardening & compliance	W10-12	TBD	PLANNED
Phase 07	Stakeholder demo & handoff	W12-14	TBD	PLANNED